

Neural Posterior Estimation and Simulation-Based Inference

A Methods Book for `extract-dvcs-gpd`

Framework methods documentation

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Purpose and claim boundary

This book explains, from first principles and down to the implemented tensor contract, how the neural network and simulation-based inference (SBI) are used in `extract-dvcs-gpd`. It is written for physicists who know the language of observables and uncertainties but need not already know neural density estimation. It is also a map from the mathematical statements to the source code and saved artifacts, so that a result can be audited rather than merely trusted.

The framework is presently a *synthetic posterior laboratory*. It learns an amortized posterior for a declared double-distribution (DD) model using exact native PARTONS simulations, synthetic experimental uncertainties, and synthetic nuisance parameters. It does not fit real measurements. The command `compare-real` is a quarantined diagnostic overlay and cannot update the posterior. “Exact” in this book means an evaluation by the native C++ bridge and PARTONS, not an analytic proof and not a Python surrogate.

The source code, configuration schemas, dependency locks, and executable validation records remain authoritative. This book describes the code as shipped in the release named on the title page; it does not silently promise methods that the code does not implement.

How to read this book

Chapters 1–4 introduce the inference problem and its probability model. Chapters 6–10 follow one simulation from an 82-component parameter vector to a conditional normalizing flow. Chapters 11–13 explain training, selection, and scientific validation. Chapters 17 and 18 are operational reading guides. The appendices give the exact parameter order, token layout, defaults, and implementation map.

A first-time user should read Chapters 1, 2, 3, 4, 8, 10, 12, 13, and 14 in that order. A reviewer can begin with the appendices and follow the file references back into the implementation.

Contents

Purpose and claim boundary	ii
How to read this book	iii
I The inverse problem	1
1 From GPD parameters to a posterior	2
1.1 What is inferred	2
1.2 The forward chain	2
1.3 Why this is a difficult inverse problem	3
2 The native forward physics simulator	5
2.1 GPD parameterization at the input scale	5
2.2 Evolution from Q_0 to each datum	5
2.3 From GPDs to Compton form factors	6
2.4 From CFFs to measured observables	6
2.5 Experimental layer and simulator interface	6
3 Bayesian simulation-based inference	7
3.1 The central idea	7
3.2 NPE, neural likelihood, and neural ratio estimation	8
3.3 What “amortized” means	8
3.4 SBI here is not a claim that no likelihood exists	8
3.5 Why NPE is not deterministic regression	9
3.6 What this framework does not use	9
4 The implemented generative model	10

4.1	Priors	10
4.2	Nuisance-dependent mean	11
4.3	Full synthetic covariance	11
5	A taxonomy of uncertainty	12
II	From exact simulations to network inputs	14
6	Reusable corpus, selections, and realizations	15
6.1	Why the data are separated into three layers	15
6.2	Corpus groups	15
6.3	Leakage-safe grouping	15
6.4	Deterministic realization	16
7	Representing a complete dataset	17
7.1	A dataset is a set of tokens	17
7.2	Why both marginal errors and whitening are present	17
7.3	What whitening does and does not do	18
8	The DeepSets neural encoder	19
8.1	Shared point network	19
8.2	Permutation-invariant pooling	19
8.3	Why DeepSets is appropriate and what it cannot guarantee	20
8.4	Fixed and variable set sizes in this implementation	20
8.5	Alternative set encoders	20
III	The conditional posterior network	22
9	Coordinates, support, and priors	23
9.1	Why bounded physical coordinates are transformed	23
9.2	Two distinct transformations	24
10	Conditional normalizing flows and the Zuko MAF	25
10.1	A density made by an invertible map	25
10.2	Masked autoregressive transforms	25

10.3	Affine MAF versus neural spline flow	26
10.4	Conditioning and posterior sampling	26
11	Training and hyperparameter optimization	28
11.1	Training and inference are different computations	28
11.2	What enters the trainer	29
11.3	Determinism and checkpoints	29
11.4	Candidate ensemble and active mixture	30
11.5	Optuna	30
IV	Validation and scientific interpretation	31
12	From posterior parameters to physics statements	32
12.1	Meaning of one posterior draw	32
12.2	Summaries and credible sets	32
12.3	Posterior predictive distribution	33
13	What validation asks	34
13.1	Optimization is not calibration	34
13.2	Outer-test negative log density	34
13.3	Repeated-coverage test	34
13.4	Simulation-based calibration and rank statistics	35
13.5	Calibration, sharpness, and simulation budget	35
13.6	Posterior predictive check	36
13.7	Exact-native reevaluation	36
13.8	What each diagnostic can and cannot establish	36
14	Identifiability, shadow directions, and model dependence	38
14.1	Null and weak directions	38
14.2	What shadow tests are actually implemented	38
14.3	D-term boundary	39
14.4	Six layers of model dependence	39
15	The conventional posterior comparison	40
15.1	Prior-bank importance sampling	40

15.2 Why ESS is decisive	40
15.3 Comparison metrics	40
16 Fresh model holdouts and real-data boundary	42
16.1 Named native-model holdouts	42
16.2 Why real data are quarantined	42
V Using and auditing the framework	43
17 Operational workflow	44
17.1 A clean campaign	44
17.2 What each expensive step computes	44
17.3 CPU and GPU roles	45
18 Reading results without overclaiming	46
18.1 A hierarchy of questions	46
18.2 Broad posterior directions	46
18.3 Correlations and marginal plots	46
18.4 Injected truth	47
18.5 Failure modes and their likely meanings	47
19 Limitations and principled extensions	48
19.1 Assumption register	48
19.2 Current limitations	49
19.3 How to extend without invalidating inference	49
A Exact posterior parameter contract	50
B Exact context contract	51
C Default neural and campaign controls	52
D Frozen validation gates	54
E Implementation map	55
F Glossary	56

References**57**

Part I

The inverse problem

Chapter 1

From GPD parameters to a posterior

1.1 What is inferred

The physics parameter vector contains independent controls for four twist-2 GPD types,

$$F \in \{H, E, \tilde{H}, \tilde{E}\}, \quad (1.1)$$

and four parton channels,

$$p \in \{u, d, s, g\}. \quad (1.2)$$

Each of the $4 \times 4 = 16$ blocks has five DD shape controls: normalization N , small- β exponent a , large- β exponent c , profile power b , and t slope B . There are therefore 80 physics coordinates. Two standard-normal experimental nuisance coordinates are appended:

$$\boldsymbol{\vartheta} = (\boldsymbol{\theta}, \eta_g, \eta_{LU}) \in \mathbb{R}^{82}. \quad (1.3)$$

Here $\boldsymbol{\theta}$ denotes the 80 physics controls, η_g a global normalization nuisance, and η_{LU} an additional normalization nuisance for the beam-spin-difference observable.

The desired object is the posterior density

$$p(\boldsymbol{\vartheta} | \mathcal{D}) = \frac{p(\mathcal{D} | \boldsymbol{\vartheta})p(\boldsymbol{\vartheta})}{p(\mathcal{D})}, \quad (1.4)$$

where $\mathcal{D}$ is one complete set of observables and their declared uncertainty model. The posterior is a joint distribution, not 82 unrelated error bars. Its correlations express degeneracies among GPD types, parton channels, shape controls, and normalizations.

1.2 The forward chain

For a parameter vector at $Q_0^2 = 1 \text{ GeV}^2$, the native calculation is conceptually

$$\boldsymbol{\theta} \longrightarrow F^p(x, \xi, t, Q_0^2) \xrightarrow{\text{APFEL++ evolution}} F^p(x, \xi, t, Q_i^2) \longrightarrow \{\mathcal{H}, \mathcal{E}, \tilde{\mathcal{H}}, \tilde{\mathcal{E}}\}_i \longrightarrow \boldsymbol{\mu}(\boldsymbol{\theta}). \quad (1.5)$$

The last vector contains the selected native DVCS observables at all declared kinematic sites. With the release default, six observables at six sites give $n = 36$ components. The Python layer does

not reproduce any GPD, CFF, or cross-section formula: it sends a versioned request to the native bridge and accepts or rejects the returned native result.

The DD model used by the bridge has the schematic form

$$F^p(x, \xi, t) = \int_0^1 d\beta \int_{-1+\beta}^{1-\beta} d\alpha \delta(x - \beta - \xi\alpha) f_{F,p}(\beta, t) \pi_{b_{F,p}}(\beta, \alpha), \quad (1.6)$$

$$f_{F,p}(\beta, t) = N_{F,p} \frac{\beta^{a_{F,p}} (1 - \beta)^{c_{F,p}}}{B(a_{F,p} + 1, c_{F,p} + 1)} e^{B_{F,p} t}. \quad (1.7)$$

The detailed native physics contract is fixed-three-flavor LO evolution in the $\overline{\text{MS}}$ scheme, a LO standard DVCS CFF module, and the native GV08 process route. Heavy-quark input, a D-term, and the $\tilde{E}$ pion pole are absent. Fixed shadow/stress settings belong to the simulator family but are not posterior coordinates.

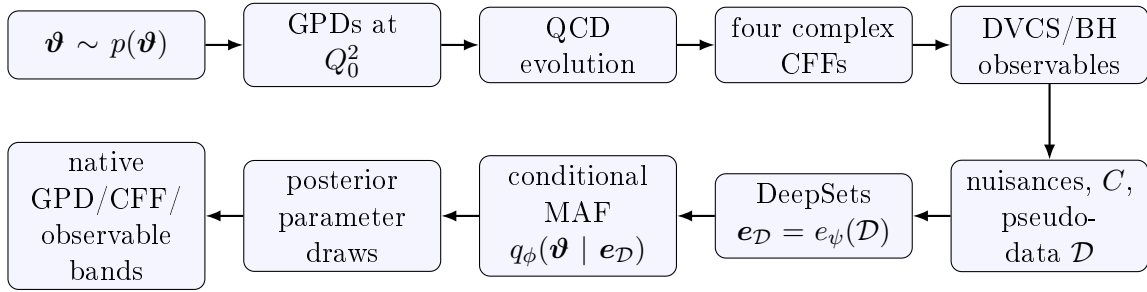


Figure 1.1: Information flow in the implemented synthetic inference chain. The top row is exact native forward physics. The lower row is statistical conditioning and posterior propagation. The final bands are produced by native reevaluation, not by treating the network as a GPD calculator.

1.3 Why this is a difficult inverse problem

The forward operator maps functions of (x, ξ, t, Q^2) to a finite collection of noisy numbers. It integrates over the partonic variable in the CFF convolution and then combines CFFs in cross sections and asymmetries. These operations discard information. Measurements cover only selected (x_B, t, Q^2, ϕ, E) points, and ξ is derived from external kinematics rather than scanned independently. Sparse coverage leaves interpolation and extrapolation directions controlled substantially by the model and prior.

The inverse is therefore many-to-one or, more commonly with finite noise, *weakly identifiable*: materially different θ can give observable vectors separated by less than experimental resolution. A change in one channel can be compensated by another; a normalization can correlate with a nuisance; small- β and large- β exponents can trade against normalization over a limited x range; and evolution couples quark singlet and gluon information. The default problem has 82 posterior coordinates but 36 measurement tokens. Counting alone does not prove non-identifiability, since nonlinear observations can constrain several directions, but it warns against assuming a unique inverse. This loss of information is the central DVCS deconvolution problem [1, 17].

Three different uncertainties must already be distinguished. *Observable uncertainty* concerns repeated measured values at fixed truth. *Parameter uncertainty* concerns which θ are compatible with

one dataset. *Functional uncertainty* is the induced distribution of $F(x, \xi, t, Q^2)$, CFFs, or derived observables after parameter samples are propagated through the physics map. Small observable error bars do not imply small functional uncertainty outside measured kinematics.

An observable-null direction is the limiting case: a deformation δF changes the GPD while its convolution and measured observables are zero, or numerically negligible, on the design. Shadow-like directions are explicitly relevant in this repository and are treated in Chapter 14. A more powerful network cannot recover a coefficient that the data do not encode.

Consequently, a single least-squares minimizer or neural point prediction is not the scientific object of interest. The relevant object is the joint distribution of GPD configurations compatible with the dataset *within the declared model class*. A scientifically honest posterior can be broad, correlated, multimodal, or prior-dominated. A posterior median need not coincide with injected truth in one noisy realization; repeated calibration, not a favorable example, tests the uncertainty statement.

Scientific caution. The neural network does not directly learn “the GPD curve.” It learns a conditional probability density over the 82 parameters. CFF and GPD bands are produced later by sending posterior parameter draws back through the native simulator.

Chapter 2

The native forward physics simulator

2.1 GPD parameterization at the input scale

The inference parameters define sixteen independent DD blocks at $Q_0^2 = 1 \text{ GeV}^2$. Equation (1.5) should therefore be read as a map from a finite-dimensional *model of functions*, not from an arbitrary element of the mathematically allowed GPD function space. The DD construction supplies support and polynomiality structure, while its Beta-like forward shape and profile impose smoothness and a particular relation between forward behavior and skewness. Independent signed normalizations do not automatically impose every proton form-factor sum rule or general positivity inequality [2].

The repository also contains native probes for a reduced conformal-moment representation and installed GK/VGG models. Those probes and post-training holdouts do not change the production NPE representation: **currently implemented inference** is the 80-control full-independent DD family. A posterior trained in that family cannot be reinterpreted as a parameterization-independent distribution over all GPDs.

2.2 Evolution from Q_0 to each datum

For datum i , the bridge invokes `PARTONS GPDEvolutionApfel` backed by `APFEL++` to compute schematically

$$F(Q_i^2) = U(Q_i^2, Q_0^2) \otimes F(Q_0^2). \quad (2.1)$$

The current contract is LO, $\overline{\text{MS}}$, fixed $n_f = 3$, with $\alpha_s(M_Z) = 0.118$. The admitted range is upward evolution from 1 to 80 GeV^2 ; backward evolution is rejected. Unpolarized and polarized GPD types use their appropriate evolution kernels. Singlet quarks and gluons mix, so a gluon input can affect evolved quarks and hence an LO quark CFF indirectly even though the direct LO gluon coefficient function vanishes.

It is the theoretical GPD that evolves. The measured value at Q_i^2 is not “moved” to Q_0^2 , which would require an extraction before the extraction and would mix experimental and theory assumptions. Each datum instead asks the same input-scale model to predict at its own physical scale.

2.3 From GPDs to Compton form factors

A GPD $F^q(x, \xi, t, Q^2)$ is a partonic function; a CFF $\mathcal{F}(\xi, t, Q^2)$ is a complex amplitude-level convolution. At a schematic factorized level,

$$\mathcal{F}(\xi, t, Q^2) = \sum_q e_q^2 \int_{-1}^1 dx C_F^q(x, \xi, Q^2/\mu_F^2, \alpha_s) F^q(x, \xi, t, \mu_F^2), \quad (2.2)$$

with analogous expressions for $H, E, \tilde{H}, \tilde{E}$. Principal value and pole structure generate real and imaginary components. The convolution is a major source of information loss: many functions can share the same few weighted integrals. The production bridge uses `DVCSFFStandard` at LO with $\mu_F^2 = \mu_R^2 = Q^2$; Python does not reimplement Eq. (2.2) or its conventions.

2.4 From CFFs to measured observables

In leptonproduction the amplitude contains calculable Bethe–Heitler (BH) and DVCS pieces,

$$|\mathcal{T}|^2 = |\mathcal{T}_{\text{BH}}|^2 + |\mathcal{T}_{\text{DVCS}}|^2 + 2 \Re(\mathcal{T}_{\text{BH}} \mathcal{T}_{\text{DVCS}}^*). \quad (2.3)$$

Beam charge, beam helicity, target polarization, and azimuthal angle select different combinations and Fourier structure of these terms. This is why the six implemented observables provide complementary, but not complete, information about real and imaginary CFF components. The native `DVCSProcessGV08` route computes the installed UU and LU cross sections and beam-charge, beam-spin, target-spin, and double-spin asymmetries with its own proton form factors and conventions. Calling it a black box in the software sense does not mean its physics is arbitrary; it means the authoritative formula and convention live in the reviewed native module.

The complete point kinematics are (x_B, t, Q^2, E, ϕ) and must also satisfy $0 < y = Q^2/(2M_p E x_B) < 1$. Cross sections use native nb/GeV⁴ before configured normalization; asymmetries are dimensionless.

2.5 Experimental layer and simulator interface

The exact native mean is only one part of the generative model. The Python workflow applies explicit normalization nuisances and draws correlated noise from the synthetic covariance of Chapter 4. Thus “simulator” in SBI means the composition

$$p(\mathcal{D} | \boldsymbol{\vartheta}) = \underbrace{p_{\text{noise}}(\mathcal{D} | \boldsymbol{\mu}_{\text{eff}}, C)}_{\text{Python statistical layer}} \circ \underbrace{\mathcal{S}_{\text{native}}(\boldsymbol{\theta})}_{\text{PARTONS/APFEL++ forward physics}}. \quad (2.4)$$

Pseudo-data uncertainty is not post-processing: it defines which datasets are probable at fixed truth and therefore changes the posterior learned by NPE.

Implementation note. `PARTONS` and the neural network are not a differentiable end-to-end model. The native C++ simulator produces saved $(\boldsymbol{\vartheta}, \mathcal{D})$ pairs. The PyTorch network differentiates only its own log density with respect to network weights. Standard NPE therefore needs simulator samples but no gradient $\partial \mathcal{D} / \partial \boldsymbol{\theta}$.

Chapter 3

Bayesian simulation-based inference

3.1 The central idea

For this project, $\boldsymbol{\theta}$ denotes the 80 input-scale GPD controls, $\boldsymbol{\vartheta} = (\boldsymbol{\theta}, \eta_g, \eta_{LU})$ the complete inferred physical vector, $\mathcal{D}$ the complete labeled DVCS dataset, and $\mathbf{e}_{\mathcal{D}} = \mathbf{e}_{\psi}(\mathcal{D})$ its learned dataset embedding. The prior $p(\boldsymbol{\vartheta})$ states which DD configurations and nuisance values enter the study. The forward simulator defines $p(\mathcal{D} \mid \boldsymbol{\vartheta})$. The evidence $p(\mathcal{D}) = \int p(\mathcal{D} \mid \boldsymbol{\vartheta})p(\boldsymbol{\vartheta})d\boldsymbol{\vartheta}$ normalizes Bayes' theorem but does not depend on a proposed parameter value.

SBI replaces repeated calls to an inconvenient likelihood inside a sampler by learning from joint draws generated by the model:

$$\boldsymbol{\vartheta}_j \sim p(\boldsymbol{\vartheta}), \quad \mathcal{D}_j \sim p(\mathcal{D} \mid \boldsymbol{\vartheta}_j). \quad (3.1)$$

If pairs $(\boldsymbol{\vartheta}_j, \mathcal{D}_j)$ can be produced, a conditional density estimator $q_{\phi}(\boldsymbol{\vartheta} \mid \mathbf{e}_{\psi}(\mathcal{D}))$ can be trained even when the simulator is non-differentiable. The trainable weights are denoted by ϕ for the flow and ψ for the jointly trained encoder. No gradient is propagated through PARTONS; gradients are needed only inside the PyTorch encoder and density estimator. This is the simulator-based perspective reviewed in [5]; the implementation uses the `sbi` toolkit [11].

The framework uses *neural posterior estimation* (NPE). It minimizes

$$\mathcal{L}(\phi, \psi) = -\mathbb{E}_{p(\boldsymbol{\vartheta}, \mathcal{D})} [\log q_{\phi, \psi}(\boldsymbol{\vartheta} \mid \mathbf{e}_{\psi}(\mathcal{D}))]. \quad (3.2)$$

For a sufficiently expressive flow, a posterior-sufficient encoder, and representative simulations, the population optimum is the true simulator posterior. To see why, condition on $\mathcal{D}$ and add and subtract the entropy of the true posterior:

$$\mathcal{L}(\phi, \psi) = \mathbb{E}_{p(\mathcal{D})} [H[p(\boldsymbol{\vartheta} \mid \mathcal{D})] + D_{\text{KL}}(p(\boldsymbol{\vartheta} \mid \mathcal{D}) \parallel q_{\phi, \psi}(\boldsymbol{\vartheta} \mid \mathbf{e}_{\psi}(\mathcal{D})))] . \quad (3.3)$$

The entropy term is independent of ϕ , and the Kullback–Leibler divergence is minimized at zero.

This identity is a population statement. A finite network, finite corpus, finite optimization, and lossy embedding need not reach the optimum. That gap is neural posterior approximation error and must be measured by held-out tests rather than assumed away.

3.2 NPE, neural likelihood, and neural ratio estimation

SBI names a family of strategies. NPE learns $q(\boldsymbol{\vartheta} \mid \mathcal{D})$ directly and is efficient when many posterior draws or many compatible observed datasets are expected. Neural likelihood estimation (NLE) learns $q(\mathcal{D} \mid \boldsymbol{\vartheta})$; one must then combine it with the prior and run MCMC or another posterior sampler for each observation. Neural ratio estimation (NRE) learns a likelihood-to-evidence or related density ratio, often through classification, and likewise needs a subsequent sampling or testing procedure.

The current use case favors NPE because the output required by downstream physics is a large posterior ensemble, the same simulator contract is to be reused across synthetic observations, and an explicit conditional sampler is valuable. This is an engineering and validation choice, not a theorem that NPE dominates NLE or NRE. If the context dimension, observation design, or tail behavior made direct posterior learning unreliable, those alternatives would deserve a controlled comparison.

3.3 What “amortized” means

A conventional sampler solves Eq. (1.4) again for each new dataset. NPE pays a large up-front simulation and training cost to learn a conditional map valid across the prior-predictive family. Once trained, conditioning on a new compatible dataset and drawing posterior samples is comparatively cheap. This reuse is called amortization.

Amortization is bounded by the training contract. A posterior is not validated for parameters outside the prior, kinematics outside the admitted domain, new observable semantics, a different covariance model, a different native bridge, or a different GPD representation. These are distribution shifts, not mere file-format changes.

Observation-specific sequential NPE could concentrate new simulations near one observed posterior. It is not implemented here. The shipped single-round method instead samples from the prior and aims to cover the full prior-predictive family; this costs more globally but supports reuse and straightforward coverage trials drawn from that same joint distribution.

3.4 SBI here is not a claim that no likelihood exists

The synthetic experimental layer deliberately uses a multivariate Gaussian noise model, so its declared likelihood can be evaluated:

$$\log p(\mathbf{y} \mid \boldsymbol{\vartheta}) = -\frac{1}{2} \mathbf{r}^T C^{-1} \mathbf{r} - \frac{1}{2} \log |C| - \frac{n}{2} \log(2\pi), \quad \mathbf{r} = \mathbf{y} - \boldsymbol{\mu}_{\text{eff}}(\boldsymbol{\vartheta}). \quad (3.4)$$

The framework nevertheless trains NPE from simulated pairs. This exercises the intended simulator-based workflow and gives rapid amortized inference. The same known likelihood is used after training for an exact-bank importance sampling comparison. NPE training itself does not minimize an observable mean-squared error or χ^2 , and it does not evaluate Eq. (3.4).

3.5 Why NPE is not deterministic regression

For a deterministic network $r(\mathcal{D})$ trained with squared error,

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}\|r(\mathcal{D}) - \boldsymbol{\vartheta}\|^2, \quad (3.5)$$

the population optimum is the conditional mean $r^*(\mathcal{D}) = \mathbb{E}[\boldsymbol{\vartheta} \mid \mathcal{D}]$. This can be useful when the posterior is unique, narrow, and approximately symmetric. It is not a posterior and supplies neither a normalized joint density nor calibrated credible regions.

Suppose a scalar ambiguity gives

$$p(\theta \mid \mathcal{D}) = \frac{1}{2}\mathcal{N}(-a, s^2) + \frac{1}{2}\mathcal{N}(a, s^2), \quad a \gg s. \quad (3.6)$$

MSE predicts zero, precisely where the posterior density is exponentially small. For a broad posterior it suppresses width; for a curved correlated posterior it returns a point that may violate the typical correlation; for a non-identifiable direction it hides that the answer remains prior-like. A sufficiently expressive conditional flow can instead represent joint width, skewness, correlations, and multiple modes. Whether the finite implemented flow actually does so is an empirical validation question.

3.6 What this framework does not use

The production path is single-round, prior-predictive NPE. It does not use neural likelihood estimation or neural ratio estimation in production, an adversarial network, approximate Bayesian computation with an acceptance threshold, MCMC over the learned density, active-learning simulation rounds, or a surrogate for PARTONS. Optuna may tune neural controls, but it does not change the physics model or use the locked outer test.

Chapter 4

The implemented generative model

4.1 Priors

Every physics coordinate is independent and uniform on its declared open interval. The two nuisance coordinates are independent standard normals:

$$N \sim \mathcal{U}(-4, 4), \quad a \sim \mathcal{U}(0.1, 1), \quad (4.1)$$

$$c \sim \mathcal{U}(2, 6), \quad b \sim \mathcal{U}(1, 4), \quad (4.2)$$

$$B \sim \mathcal{U}(0, 2) \text{ GeV}^{-2}, \quad \eta_g, \eta_{LU} \sim \mathcal{N}(0, 1). \quad (4.3)$$

This factorization is a modeling decision. It should not be read as a claim that real GPD controls are physically independent.

The prior is simultaneously a Bayesian assumption and the NPE training distribution. Outside its support the trained posterior has no inferential meaning. A narrow prior can exclude physically plausible solutions and make a posterior look artificially precise. An extremely broad prior spends a finite simulation budget on extreme or invalid regions and can make an 82-dimensional conditional density too sparse to learn. “Broad” is therefore not automatically conservative.

A prior-predictive check draws

$$\boldsymbol{\vartheta}^{(s)} \sim p(\boldsymbol{\vartheta}), \quad \mathcal{D}^{(s)} \sim p(\mathcal{D} \mid \boldsymbol{\vartheta}^{(s)}) \quad (4.4)$$

before training and compares the resulting observable ranges, validity rate, and correlations with the intended experiment. It asks whether the study even assigns probability to datasets of the relevant scale without being dominated by pathological simulations. The corpus invalid map and native range checks supply part of this evidence; a complete scientific campaign should also inspect prior-predictive plots. The public release does not currently expose an automated prior-predictive acceptance gate.

Prior sensitivity is distinct from neural-seed robustness. Repeating an analysis under defensible alternative supports or correlated priors changes the Bayesian question and requires a newly materialized/trained inference contract, though a compatible noise-free native corpus may sometimes be reselected. Comparing such analyses reveals how much a conclusion is driven by prior information.

4.2 Nuisance-dependent mean

Let $\mu_i(\boldsymbol{\theta})$ be the normalized native prediction. The effective mean is

$$\mu_{\text{eff},i}(\boldsymbol{\vartheta}) = \mu_i(\boldsymbol{\theta}) (1 + r_g \eta_g + r_{LU,i} \eta_{LU}), \quad (4.5)$$

with $r_g = 0.02$ for every point. The response $r_{LU,i} = 0.03$ only for the beam-spin-difference observable `DVCSCrossSectionDifferenceLUMinus`; otherwise it is zero. The nuisance action is linear and fractional. The nuisance values are inferred jointly with the physics controls.

4.3 Full synthetic covariance

The covariance is fixed at the declared truth prediction $\boldsymbol{\mu}^{\text{ref}}$. Define

$$\sigma_i = f_0 + f_r |\mu_i^{\text{ref}}|, \quad (4.6)$$

$$\ell_i = f_{\text{loc}} \max(|\mu_i^{\text{ref}}|, f_0), \quad (4.7)$$

$$K_{ij} = \exp(-|i - j|/L), \quad (4.8)$$

$$s_i = f_\phi \mu_i^{\text{ref}} \sin \phi_i. \quad (4.9)$$

The implemented covariance is

$$C_{ij} = \delta_{ij} \sigma_i^2 + \ell_i \ell_j K_{ij} + s_i s_j, \quad (4.10)$$

where $f_r = 0.05$, $f_0 = 0.01$, $f_{\text{loc}} = 0.015$, $L = 1.5$, and $f_\phi = 0.01$ in normalized-observable units. It is explicitly symmetrized and required to be positive definite. No diagonal approximation or numerical regularizer is silently substituted.

One training realization is then

$$\mathbf{y} = \boldsymbol{\mu}_{\text{eff}}(\boldsymbol{\vartheta}) + L_C \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I), \quad L_C L_C^T = C. \quad (4.11)$$

The Cholesky factor L_C is used to draw correlated noise. The context uses a different, symmetric whitening operation described in Chapter 7; the two operations serve different purposes.

Scientific caution. Equation (4.10) is a synthetic study design, not an experimental covariance claim. Real inference remains disabled until observable conventions, units, systematics, correlations, and nuisance assignments are audited dataset by dataset.

Chapter 5

A taxonomy of uncertainty

The word “uncertainty” is dangerous unless its source is named. The current posterior automatically propagates only randomness and unknowns that appear in the declared generative model. Other uncertainty sources require separate models or sensitivity studies.

Source	Meaning	Status in the current framework
Experimental statistical	Repeat-to-repeat measurement fluctuations at fixed truth	Represented by the diagonal/floor and correlated synthetic Gaussian covariance; not yet a real experiment model.
Experimental systematic	Shared calibration, normalization, acceptance, or shape effects	Two normalization nuisances and two synthetic correlated covariance sources are represented. Real dataset systematics are not.
Parameter/posterior	Uncertainty in the 82 inferred coordinates conditional on one dataset and model	The direct target of NPE.
Functional	Uncertainty in GPDs, CFFs, or observables induced by posterior coordinates	Represented only after exact native reevaluation of joint posterior draws.
Prior sensitivity	Change caused by alternative defensible priors	Not marginalized automatically; requires separate campaigns.
Model and parameterization	Change caused by DD versus conformal or other GPD families, omitted D-term, poles, or constraints	Outside one posterior; native probes and named holdouts provide limited stress evidence.
Perturbative theory	Missing orders, scheme/scale variation, flavor thresholds	Not included; LO and fixed scheme/flavor choices are frozen.
Numerical simulator	Quadrature, evolution discretization, floating-point, or native failures	Validity, hashes, and regression tests are recorded, but no numerical-error nuisance is integrated into the posterior.

Source	Meaning	Status in the current framework
Neural approximation	Difference between $q_{\phi,\psi}$ and the target posterior	Diagnosed by validation NLL, coverage, conventional comparison, exact reevaluation, and seed ensemble; never eliminated by definition.
Finite simulation/training	Monte Carlo sparsity and optimization variability	Partly probed by profiles, held-out groups, seeds, and coverage precision; simulation-budget convergence is still a campaign responsibility.

Thus a 90% neural credible band means approximately “90% posterior mass under the declared prior, DD family, LO simulator, synthetic uncertainty model, and learned approximation.” It is not a 90% statement after marginalizing every experimental and theoretical uncertainty relevant to nature.

Part II

From exact simulations to network inputs

Chapter 6

Reusable corpus, selections, and realizations

6.1 Why the data are separated into three layers

Native simulation is the expensive part, whereas nuisance and noise draws are cheap. The framework therefore separates:

1. the *corpus*: immutable, noise-free native results;
2. the *selection*: immutable group assignments; and
3. the *realization*: project-specific nuisance/noise tensors and neural contexts.

This permits the same exact native corpus to support later neural studies without pretending that a changed simulator or kinematic design is compatible.

6.2 Corpus groups

One corpus group is one accepted 80-component physics vector evaluated at all declared kinematics. A shard stores the parameter vectors, complex CFFs, and native observables, together with request/response evidence and hashes. Invalid native draws are recorded and replaced by later prior draws; they are not imputed and no surrogate prediction is inserted.

The corpus contains no nuisance draws, covariance noise, network architecture, epoch count, or train/test role. Consequently, changing only those items does not require repeating PARTONS. Changing the native bridge identity, DD physics, evolution, kinematics, observable semantics, or another field in the corpus identity does require a new or explicitly extended verified corpus.

6.3 Leakage-safe grouping

Several noisy replicas can be generated from one exact physics vector. A random row split would leak physics: the network could see one noisy version of a vector during training and another during validation or testing. The framework instead splits native parameter *groups* first, then assigns every replica of a group to the same role.

With the default held-out fraction and validation fraction, approximately 72% of native groups are training groups, 8% are internal-validation groups, and 20% are locked outer-test groups. Rounding is explicit for finite corpus sizes. The three lists must be nonempty, disjoint, in range, and collectively complete.

The internal validation set controls early stopping, Optuna’s objective, and ensemble-member selection. The outer test is never appended to the NPE trainer and is evaluated only after the candidates have been ranked.

6.4 Deterministic realization

For each selected native group and each configured replica, the realization draws (η_g, η_{LU}) , constructs Eq. (4.5), draws Eq. (4.11), and builds the context. Seeds are frozen in the configuration. The parameter-group and replica identities therefore remain auditable even though the network consumes flattened rows.

The quick profile requests 2,048 native groups and two replicas per group; the validation profile requests 4,096 groups and four replicas. These are compute defaults, not a theorem that their density is sufficient in 82 dimensions. A very large corpus still has to fit in the current materialized tensor path; lazy streaming is not implemented.

Chapter 7

Representing a complete dataset

7.1 A dataset is a set of tokens

Each observation becomes an 18-component token $\mathbf{x}_i$. With six observables at six sites the default dataset has 36 tokens. The order in which those points happen to be stored should not change the inferred posterior, so the encoder is designed to be permutation invariant.

The first six features encode kinematics:

$$\left(\frac{x_B}{0.3}, \frac{-t}{0.2 \text{ GeV}^2}, \frac{Q^2}{Q_0^2}, \frac{E}{12 \text{ GeV}}, \sin \phi, \cos \phi \right), \quad (7.1)$$

where $Q_0^2 = 1 \text{ GeV}^2$. The next six are a one-hot observable type. The remaining six give the observed normalized value, marginal standard deviation, symmetric-whitened value divided by 10, global response, LU response, and a point mask equal to one. Appendix B freezes the exact wire order.

Nine dataset-level contract features are appended after all flattened tokens. They currently equal one and assert the representation, input scale, LO coefficient route, configured evolution, native datum- Q^2 evolution, full covariance, four-GPD content, native six-observable route, and model mask. They are semantic guards and extension points, not learned measurements.

7.2 Why both marginal errors and whitening are present

The diagonal feature $\sqrt{C_{ii}}$ tells each token its marginal scale, but diagonal scales alone discard correlation. Let

$$C = V\Lambda V^T \quad (7.2)$$

be the eigendecomposition of the positive-definite covariance. The context computes

$$\mathbf{y}_w = C^{-1/2} \mathbf{y} = V\Lambda^{-1/2} V^T \mathbf{y}, \quad (7.3)$$

and gives token i the component $y_{w,i}/10$.

A triangular Cholesky whitening depends on which point is listed first. In contrast, if a permutation matrix P reorders both data and covariance, then

$$(PCP^T)^{-1/2} P\mathbf{y} = PC^{-1/2} \mathbf{y}. \quad (7.4)$$

Thus symmetric whitening is permutation equivariant: reordered inputs produce reordered whitened token features. The subsequent set pooling then removes the order.

7.3 What whitening does and does not do

Whitening exposes the known full-covariance geometry to the encoder. It does not turn the NPE objective into a Gaussian likelihood, and it is not a sufficient statistic guaranteed to preserve every aspect of the dataset. The raw observed value, covariance responses, kinematics, and observable type remain available to the network. The factor of 10 is a fixed feature scale, not an uncertainty inflation.

Chapter 8

The DeepSets neural encoder

8.1 Shared point network

The same multilayer perceptron h_ψ is applied to every token:

$$\mathbf{e}_i = h_\psi(\mathbf{x}_i). \quad (8.1)$$

At the release default, the point network has three 128-wide linear layers with SiLU nonlinearities. An optional LayerNorm immediately after the first activation exists but is disabled by default. Sharing weights means that a beam-spin token is processed by the same rule regardless of its list index; the one-hot features tell the rule which observable it represents.

The SiLU activation is

$$\text{SiLU}(u) = u \sigma(u), \quad \sigma(u) = \frac{1}{1 + e^{-u}}. \quad (8.2)$$

It supplies smooth nonlinearity. A linear network alone could only form linear summaries and would not express the nonlinear relation between kinematics, observable class, values, and covariance information.

8.2 Permutation-invariant pooling

The point embeddings are mean pooled:

$$\bar{\mathbf{e}} = \frac{1}{n} \sum_{i=1}^n \mathbf{e}_i. \quad (8.3)$$

For any permutation π , the sum and therefore the mean are unchanged. This is the implemented DeepSets invariance. Mean rather than sum pooling also keeps the representation scale from growing linearly with token count, although a changed observable subset still changes the experiment contract and requires compatible training.

The mean-pooled vector is concatenated with the nine global features and sent through a dataset-level multilayer perceptron g_ρ :

$$\mathbf{e}_\mathcal{D} = g_\rho([\bar{\mathbf{e}}, \mathbf{m}]) \in \mathbb{R}^{96}. \quad (8.4)$$

The default dataset network has two layers with hidden width 128 and SiLU between them. The resulting 96-component $\mathbf{e}_\mathcal{D}$ conditions every autoregressive transform in the posterior flow.

8.3 Why DeepSets is appropriate and what it cannot guarantee

DeepSets encodes the physical idea that a dataset is a collection of labeled measurements, not an image whose neighboring array cells have universal meaning [7]. It also makes an input-order regression test mathematically possible. It does not guarantee that the 96-component summary is sufficient, that finite training finds the optimal summary, or that unseen kinematic designs are calibrated. Those questions are answered empirically through held-out density, coverage, and predictive tests.

Flattening the 36 tokens directly into an MLP would attach physical meaning to array positions: exchanging two equally valid point records would change the output unless the network learned every relevant permutation. Processing each point with an independent posterior would fail in the opposite direction, because a parameter vector is constrained by the *joint* pattern across kinematics and observables. The shared map plus pooling is a compromise: it is order-free while producing one dataset-level context.

The covariance is not modeled by pairwise message passing. Instead, its effect enters each token through marginal scale and the globally computed symmetric-whitened observation. This preserves permutation equivariance but may not expose every useful pairwise structure as explicitly as attention or a graph encoder would.

8.4 Fixed and variable set sizes in this implementation

DeepSets as a mathematical family can accept variable n . The current `DeepSetsDatasetEncoder`, however, is constructed with `token_count` equal to the configured number of kinematics times the configured number of observables, checks the flattened width exactly, and mean-pools that fixed reshape. The point mask is presently one for every token. An admitted observable subset creates a correspondingly different encoder shape and requires compatible training; one checkpoint does not accept arbitrary missing points. Variable-cardinality masking is therefore a **possible future study**, not a released capability.

8.5 Alternative set encoders

DeepSets is the **current baseline**, not an empirical optimum. Useful controlled alternatives include:

Set Transformer or self-attention

Each point can attend directly to every other point, allowing learned kinematic and cross-observable interactions before pooling. This can help when relations between pairs carry information, at $O(n^2)$ attention cost and with greater simulation demand [13].

Graph neural network

Nodes are measurement points and edges can encode shared dataset, observable, systematic group, or kinematic proximity. This makes a chosen relational inductive bias explicit, but the graph construction itself becomes a scientific assumption.

Fixed-grid MLP or convolutional encoder

Efficient when every experiment lies on the same dense regular grid. It is poorly matched to irregular DVCS coverage and introduces interpolation or padding conventions.

Recurrent encoder

Consumes a sequence and can handle varying length, but order dependence must be removed by canonical sorting or augmentation; neither is physically natural for an unordered dataset.

Learned summary-statistic network

Any of the above can be trained to compress $\mathcal{D}$. The central risk is information loss: a summary useful for one parameter combination may erase a weak direction. NPE’s joint log-density objective trains the current summary end-to-end, but does not prove sufficiency.

Comparisons should hold the simulator bank and split fixed, then examine held-out NLL, coverage, predictive checks, seed robustness, and compute. A lower training loss alone would not establish a better scientific encoder.

Part III

The conditional posterior network

Chapter 9

Coordinates, support, and priors

9.1 Why bounded physical coordinates are transformed

Normalizing flows are easiest to construct on unbounded Euclidean space. For each bounded physics coordinate $\theta_k \in (\ell_k, u_k)$, the framework uses the analytic probability integral transform

$$z_k = \Phi^{-1} \left(\frac{\theta_k - \ell_k}{u_k - \ell_k} \right), \quad (9.1)$$

where Φ is the standard-normal cumulative distribution function. The inverse is

$$\theta_k = \ell_k + (u_k - \ell_k)\Phi(z_k). \quad (9.2)$$

Uniform physical priors therefore become exact independent standard-normal latent priors. The two nuisance parameters already have that prior and are left unchanged. In latent coordinates,

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_{82}). \quad (9.3)$$

The physical endpoints map to infinity and have probability zero under the continuous uniform prior, so inputs must lie strictly inside their bounds. Posterior samples transformed by Eq. (9.2) remain in physical support up to floating-point behavior.

For densities, a reparameterization carries a Jacobian. In one bounded coordinate, writing $r = (\theta - \ell)/(u - \ell)$ and $z = \Phi^{-1}(r)$,

$$\left| \frac{dz}{d\theta} \right| = \frac{1}{(u - \ell) \varphi_{\mathbf{N}}(z)}, \quad (9.4)$$

where $\varphi_{\mathbf{N}}$ is the standard-normal density. Therefore a physical-coordinate density would satisfy

$$q_{\theta}(\boldsymbol{\theta} \mid \mathcal{D}) = q_z(h(\boldsymbol{\theta}) \mid \mathcal{D}) \left| \det \frac{\partial h}{\partial \boldsymbol{\theta}} \right|. \quad (9.5)$$

The implementation trains and reports NLL in latent $\mathbf{z}$ coordinates; it does not claim those numerical log densities are physical-coordinate densities. Sampling needs no explicit Jacobian multiplication: drawing $\mathbf{z}$ and applying the deterministic inverse automatically pushes the probability measure into physical space. The Jacobian has already entered the mathematical relation between the two densities.

9.2 Two distinct transformations

It is useful not to confuse:

- the analytic probit, which maps physical posterior coordinates to latent coordinates and exactly represents bounded support; and
- the estimator’s configured independent $\mathbf{z}$ standardization, learned from the training batch by `sbi` before the Zuko flow.

Because the analytic latent prior is already standard normal, the second operation should be close to identity for a representative corpus, but it is still part of the saved estimator. Context standardization is disabled; feature scaling is explicit in the token contract.

This support matching improves optimization because the flow does not need to learn sharp hard walls at five repeated parameter bounds. It also prevents ordinary posterior samples from wandering outside configured support. It does not enforce additional physics constraints absent from those scalar bounds, such as positivity or form-factor sum rules.

Chapter 10

Conditional normalizing flows and the Zuko MAF

10.1 A density made by an invertible map

A normalizing flow transforms a simple base variable $\mathbf{u} \sim p_0(\mathbf{u})$ into a complex variable $\mathbf{z}$ using an invertible, context-dependent map

$$\mathbf{z} = T_\phi(\mathbf{u}; \mathbf{e}_\mathcal{D}). \quad (10.1)$$

The change-of-variables formula gives an exactly evaluable conditional density:

$$\log q_\phi(\mathbf{z} | \mathbf{e}_\mathcal{D}) = \log p_0(\mathbf{u}) - \log \left| \det \frac{\partial T_\phi(\mathbf{u}; \mathbf{e}_\mathcal{D})}{\partial \mathbf{u}} \right|, \quad \mathbf{u} = T_\phi^{-1}(\mathbf{z}; \mathbf{e}_\mathcal{D}). \quad (10.2)$$

This is a normalized joint density in 82 dimensions, not a collection of Gaussian regressions. Its samples can contain nonlinear correlations and non-Gaussian marginal shapes.

A single multivariate Gaussian would be fully characterized by one mean and covariance. It cannot represent bounded-coordinate skewness after returning to physical space, separated modes, heavy or asymmetric tails, or curved degeneracy manifolds. A flow retains an exactly normalized, sampleable density while composing invertible nonlinear maps. In the common inverse-map notation $\mathbf{u} = f_\phi^{-1}(\mathbf{z}; \mathbf{e}_\mathcal{D})$,

$$q_\phi(\mathbf{z} | \mathbf{e}_\mathcal{D}) = q_0(\mathbf{u}) \left| \det \frac{\partial f_\phi^{-1}(\mathbf{z}; \mathbf{e}_\mathcal{D})}{\partial \mathbf{z}} \right|. \quad (10.3)$$

The base q_0 supplies simple random draws; the invertible transform reshapes them; the determinant accounts for local expansion or compression of volume; and the context $\mathbf{e}_\mathcal{D}$ makes a different posterior shape for each dataset.

10.2 Masked autoregressive transforms

The configured model name is `zuko_maf`: a Zuko masked autoregressive flow (MAF) built by `sbi`. An autoregressive transform gives coordinate k parameters that depend only on earlier coordinates

in a chosen ordering and on the dataset embedding:

$$z_k = a_k(\mathbf{z}_{<k}, \mathbf{e}_{\mathcal{D}})u_k + b_k(\mathbf{z}_{<k}, \mathbf{e}_{\mathcal{D}}), \quad a_k > 0. \quad (10.4)$$

Masks in the neural hypernetwork enforce this dependency. The Jacobian is triangular, so its log determinant is the sum of diagonal log scales. The Zuko base density is an 82-dimensional diagonal standard normal.

The release stacks six autoregressive transforms. With the default `randperm=False` inherited through `sbi`, Zuko alternates forward and reversed coordinate order between transforms. Each transform’s masked hypernetwork is conditioned on the 96-component DeepSets embedding. Under the pinned builder, the scalar `hidden_features=96` and `num_transforms=6` become six ReLU hidden layers of width 96 in *each* masked conditioner, followed by 164 outputs: a shift and scale for each of 82 coordinates. This depth is a consequence of the current `sbi` 0.26.1 Zuko builder and is part of the effective architecture, even though the JSON exposes the width and transform count rather than a separate conditioner depth. The triangular-density construction follows [8]; normalizing-flow principles are reviewed in [10].

Implementation note. Although the JSON contract retains a `num_bins` field with default 8, `sbi` 0.26.1 explicitly removes `num_bins` when building a Zuko MAF. The effective univariate transform in Zuko 1.6.0 is monotonic affine, not a neural spline. Consequently `num_bins` has no effect on the current `zuko_maf` estimator. It is a compatibility field, and Optuna trials that differ only in this field do not represent different effective MAF architectures.

10.3 Affine MAF versus neural spline flow

A neural spline flow (NSF) replaces each monotonic affine scalar map by a piecewise rational-quadratic monotonic spline whose bin widths, heights, and derivatives are predicted by a conditioner. This gives more local curvature per transform and can represent complicated one-dimensional conditional shapes with fewer stacked layers. Its change-of-variables principle and triangular autoregressive Jacobian are the same, but bin count and tail bound become effective hyperparameters [9].

The current estimator is *not* an NSF. The pinned `posterior_nn(model="zuko_maf")` path removes spline-only arguments and constructs Zuko monotonic affine autoregressive transforms. An NSF is a **possible future comparison** that would need a new configuration, checkpoint contract, tuning study, and calibration campaign. More flexible scalar transforms can reduce approximation bias, but can also increase optimization variance and simulation requirements; they are not automatically superior.

10.4 Conditioning and posterior sampling

For an observed compatible dataset, the same context builder and DeepSets encoder produce $\mathbf{e}_{\text{obs}}$. Sampling performs

$$\mathbf{u}^{(s)} \sim p_0, \quad \mathbf{z}^{(s)} = T_\phi(\mathbf{u}^{(s)}; \mathbf{e}_{\text{obs}}), \quad \boldsymbol{\vartheta}^{(s)} = h^{-1}(\mathbf{z}^{(s)}), \quad (10.5)$$

where h^{-1} is Eq. (9.2) for physics coordinates and the identity for nuisances. Saved checkpoints contain state dictionaries rather than pickled arbitrary Python objects; loading reconstructs the architecture from configuration and applies the state strictly.

The code objects have a direct mathematical interpretation. `DeepSetsDatasetEncoder` implements $e_\psi(\mathcal{D})$. `posterior_nn` is the `sbi` factory that combines this encoder with the conditional Zuko MAF. `stage10_prior` is the factorized latent $\mathcal{N}(0, I_{82})$ prior induced by the physical transform. `DirectPosterior` wraps the trained estimator and prior to provide conditional sampling and log probability in latent coordinates. Finally, `stage10_latent_to_physical` returns joint samples to the parameter contract used by the native simulator. `DeepSets` summarizes the data; the MAF represents the posterior. Neither component alone performs both jobs.

Chapter 11

Training and hyperparameter optimization

11.1 Training and inference are different computations

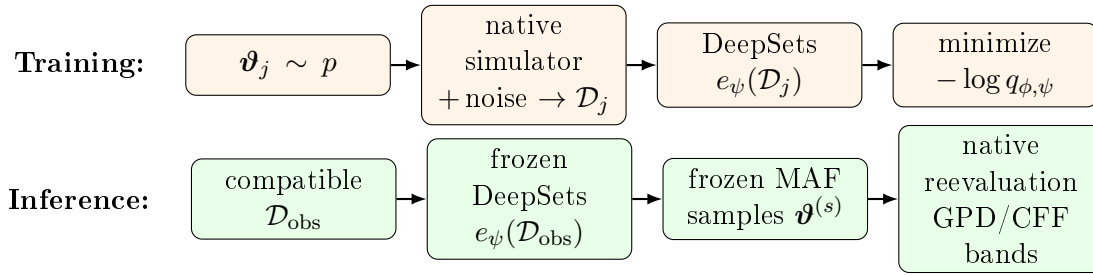


Figure 11.1: Training uses many simulated parameter–dataset pairs to optimize network weights. Inference freezes those weights, conditions on one dataset, and produces parameter samples. Native reevaluation then maps those samples to physics functions. No experimental datum is back-propagated through PARTONS.

The actual workflow can be followed step by step:

1. Draw an 80-component $\boldsymbol{\theta}_j$ from the bounded physics prior. This defines which native inverse problems the corpus will span.
2. Construct the corresponding DD GPD at Q_0^2 . Native validity checks reject rather than impute unusable physics draws.
3. Evolve that one input-scale configuration to every requested Q_i^2 , ensuring one coherent GPD explains all scales.
4. Compute four complex CFFs and then all configured native observables at the exact experimental kinematics.
5. Store the noise-free result in a reusable corpus. This prevents an architecture study from needlessly repeating native physics.
6. Freeze group-disjoint training, internal-validation, and outer-test ownership before adding replicas.
7. For every replica, draw the two nuisance coordinates, apply their normalization responses, and draw full-covariance noise, producing $\mathcal{D}_j \sim p(\mathcal{D} \mid \boldsymbol{\vartheta}_j)$.

8. Convert $\boldsymbol{\vartheta}_j$ to unconstrained $\mathbf{z}_j$ and construct the 18-feature point tokens and nine global features.
9. DeepSets maps the complete dataset to $\mathbf{e}_j$; the conditional MAF assigns a joint density to $\mathbf{z}_j$.
10. Minibatch gradient descent updates ψ and ϕ to reduce Eq. (3.2); internal validation controls stopping and member selection.
11. Freeze the selected ensemble. Outer-test, named-model, and real diagnostic outputs cannot update its weights.
12. Build the same context for a compatible observed pseudo-dataset and draw $\mathbf{z}^{(s)} \sim q_{\phi,\psi}(\mathbf{z} \mid \mathbf{e}_{\text{obs}})$.
13. Invert the probit to obtain physical $\boldsymbol{\vartheta}^{(s)}$ and send the 80 physics coordinates through the native bridge.
14. Summarize the resulting joint parameter, GPD, CFF, and observable ensembles with credible intervals, correlations, and posterior-predictive checks.

Steps 1–10 are amortized training. Steps 11–14 are conditional inference and scientific interpretation. Repeating only the latter is cheap relative to rebuilding a native corpus, provided the new dataset is inside the frozen contract.

11.2 What enters the trainer

The trainer receives only latent parameter/context pairs from training and internal-validation groups. Outer-test rows are excluded before simulations are appended to the trainer. The custom grouped-validation adapter overrides `sbi`’s data loaders so that the frozen group membership is used instead of a new random row split. In the source these two interfaces are named

```
append_simulations
GroupedValidationNPE
```

The adapter does not replace `sbi`’s NPE objective, optimizer, density estimator, or training loop. For a minibatch B , the empirical objective is

$$\hat{\mathcal{L}}_B(\phi, \psi) = -\frac{1}{|B|} \sum_{j \in B} \log q_{\phi,\psi}(\mathbf{z}_j \mid e_{\psi}(\mathcal{D}_j)). \quad (11.1)$$

The default batch size is 256 and the learning rate is 5×10^{-4} . The quick profile permits at most 100 epochs and stops after 20 epochs without validation improvement; validation permits at most 200 epochs with patience 25. The best validation state is returned by the `sbi` training loop.

11.3 Determinism and checkpoints

Each candidate member has a frozen seed. Python/NumPy and Torch seeds are set; deterministic Torch algorithms are requested; CPU threads and resolved device are recorded. The output stores the state hash, input-manifest hash, software versions, loss histories, elapsed time, warnings, parameter count, and CUDA memory metrics. Restart logic refuses incomplete or mismatched state/metric pairs instead of silently mixing experiments.

Deterministic algorithms improve repeatability under the same supported software/hardware path. They do not imply that CPU and GPU arithmetic is bitwise identical, or that different library versions reproduce identical weights.

11.4 Candidate ensemble and active mixture

The quick profile trains two candidate seeds and retains both. The validation profile trains five candidates and retains the three with the lowest grouped internal-validation negative log density. Only after this ranking are active members scored on the outer test. Thus the test cannot choose a seed.

Posterior draws are allocated as evenly as possible across active members and concatenated. This is an equal-weight empirical mixture. If M members are active, its density conceptually is

$$q_{\text{ens}}(\mathbf{z} | \mathbf{e}_{\mathcal{D}}) = \frac{1}{M} \sum_{m=1}^M q_{\phi_m}(\mathbf{z} | \mathbf{e}_{\mathcal{D}}), \quad (11.2)$$

although the framework’s downstream operations use mixture samples rather than explicitly evaluating this mixture density. Independent ensemble members can be sharded across visible GPUs; one member itself is not split across GPUs.

11.5 Optuna

Currently implemented: the optional optimizer uses a seeded TPE sampler and a persistent SQLite study. Its objective is the best grouped-validation negative log density. It searches point and dataset widths, embedding width, flow width, transform count, batch size, and learning rate. The declared `num_bins` search is ineffective for the current Zuko MAF for the reason stated in Chapter 10. A trial uses one optimization seed and cannot evaluate the outer test.

The MedianPruner decision is recorded after training histories are returned; this release does not claim mid-epoch compute savings. The recommended network must be copied into a *new* project, followed by one clean generation/selection/training/validation campaign. Repeatedly consulting an outer test turns it into validation data, regardless of what the file is called.

Current baseline, not an optimum: three 128-wide SiLU point layers, two 128/96 dataset layers with one SiLU, mean pooling, a 96-dimensional embedding, six affine autoregressive transforms whose masked conditioners each have six ReLU hidden layers of width 96, batch size 256, and learning rate 5×10^{-4} . Point-layer count, dataset-layer count, activation, LayerNorm switch, and pooling operator are not all searched by the current public Optuna space. Appendix C gives the exact defaults.

Possible future study: compare layer counts, LayerNorm, alternative pooling, attention encoders, spline or other flow families, transform depth, and larger simulation budgets. Selection should use validation NLL as one criterion, then predeclared coverage, posterior predictive behavior, robustness across seeds, native validity, and computational cost on a clean campaign. Calibration statistics are noisy, so repeatedly tuning against the same finite coverage set would overfit that diagnostic. Training loss alone rewards fit to seen pairs and cannot establish uncertainty quality.

Part IV

Validation and scientific interpretation

Chapter 12

From posterior parameters to physics statements

12.1 Meaning of one posterior draw

A draw

$$\boldsymbol{\vartheta}^{(s)} = (\boldsymbol{\theta}^{(s)}, \eta_g^{(s)}, \eta_{LU}^{(s)}) \sim q_{\text{ens}}(\boldsymbol{\vartheta} \mid \mathcal{D}_{\text{obs}}) \quad (12.1)$$

is one complete, internally correlated hypothesis about all 80 DD controls and two nuisances after conditioning on the dataset. Coordinates from different draws must never be mixed independently: doing so destroys correlations that may be essential to reproduce the observables.

The native parameterization turns the physics part into a function at the input scale,

$$\boldsymbol{\theta}^{(s)} \longmapsto F^{p,(s)}(x, \xi, t, Q_0^2), \quad (12.2)$$

and evolution gives

$$F^{p,(s)}(x, \xi, t, Q^2) = U(Q^2, Q_0^2) \otimes F^{p,(s)}(x, \xi, t, Q_0^2). \quad (12.3)$$

Repeating this map for joint draws produces a posterior ensemble of GPD functions. Repeating the CFF convolution and process calculation gives CFF and observable ensembles. The network provides parameter probability; the native physics map provides the meaning of those parameters.

In the current public workflow, $\mathcal{D}_{\text{obs}}$ in Eq. (12.1) is the generated project pseudo-dataset or a held-out synthetic/native-model diagnostic with the frozen context contract. Supplying real measured values does *not* trigger posterior inference: real-data fitting is disabled, and **compare-real** is a read-only overlay. A future real-data path would reuse the trained-inverse logic only after the real observable and uncertainty model has been audited and included in the training simulator.

12.2 Summaries and credible sets

Marginal histograms answer questions about one coordinate after integrating over all others. Pair plots or correlation matrices reveal compensation and degeneracy. A pointwise central 90% GPD band at phase-space coordinate r uses the 5th and 95th percentiles of $\{F^{(s)}(r)\}_{s=1}^S$. This is not a

simultaneous band over all r , and it is not obtained by inserting marginal parameter endpoints into a nonlinear formula.

The same distinction holds for CFFs. Each complex CFF has real and imaginary ensembles at every kinematic point; credible regions may be strongly correlated across components and kinematics. Derived quantities should be computed for every joint sample first and summarized second. “Propagating errors” by a local linear covariance approximation is unnecessary when exact posterior draws and the native forward model are available.

Posterior summaries are conditional on the model class. A narrow band can mean that the data constrain the DD coordinates inside this family while an omitted D-term, alternative skewness construction, perturbative order, or experimental systematic would widen the scientifically relevant uncertainty.

12.3 Posterior predictive distribution

The posterior over parameters induces the observable distribution

$$p(\mathcal{D}_{\text{rep}} \mid \mathcal{D}_{\text{obs}}) = \int p(\mathcal{D}_{\text{rep}} \mid \boldsymbol{\vartheta}) p(\boldsymbol{\vartheta} \mid \mathcal{D}_{\text{obs}}) d\boldsymbol{\vartheta}. \quad (12.4)$$

Operationally,

$$\boldsymbol{\vartheta}^{(s)} \sim q_{\text{ens}}(\cdot \mid \mathcal{D}_{\text{obs}}), \quad \mathcal{D}_{\text{rep}}^{(s)} \sim p(\mathcal{D} \mid \boldsymbol{\vartheta}^{(s)}). \quad (12.5)$$

Replicated cross sections and asymmetries are compared with the conditioned data using the same nuisance and covariance semantics. A narrow parameter posterior is scientifically unhelpful if its replicated observables fail to cover the data: that combination signals approximation error, model misspecification, an inconsistent uncertainty model, or all three.

Posterior predictive agreement is necessary but not sufficient. A flexible or non-identifiable model may reproduce observables while leaving GPDs broad, and a misspecified model can sometimes fit measured points yet extrapolate poorly. Coverage at known synthetic truths and fresh holdouts address those different questions.

Chapter 13

What validation asks

13.1 Optimization is not calibration

Training answers: “Can the flow assign high conditional density to held-out simulator parameters?” Scientific validation asks additional questions: Does it cover truths at the stated frequency? Does it generate datasets like the observed one? Does it agree with an independent posterior calculation when that calculation is reliable? Are sampled physics points valid under exact native reevaluation? Does a frozen estimator behave sensibly on fresh named native models?

No single scalar answers all of these. The framework therefore keeps role boundaries and records separate gates.

13.2 Outer-test negative log density

After validation-only member selection, the selected estimators score the locked outer-test pairs. The recorded diagnostic compares mean outer-test negative log density with best internal-validation negative log density. A large positive gap indicates distribution-generalization or overfitting problems. Absolute NLL is architecture- and coordinate-dependent, so it is not a universal physics goodness-of-fit statistic.

13.3 Repeated-coverage test

For held-out synthetic trial r and credibility level α , equal-tailed intervals are computed from posterior samples for every coordinate. The empirical coverage for coordinate k is

$$\hat{c}_{k,\alpha} = \frac{1}{R} \sum_{r=1}^R \mathbf{1}\{\vartheta_{rk}^* \in [Q_{(1-\alpha)/2}, Q_{(1+\alpha)/2}]\}. \quad (13.1)$$

The reference binomial standard error is

$$\text{SE}_\alpha = \sqrt{\frac{\alpha(1-\alpha)}{R}}. \quad (13.2)$$

The framework evaluates levels 0.5, 0.8, and 0.9, and gates the largest coordinate/level deviation measured in these standard errors. Quick uses at most 20 trials and validation at most 80. Such finite tests have substantial Monte Carlo resolution; passing is evidence for that campaign, not proof of universal calibration.

This is coverage testing closely related to simulation-based calibration, but the saved diagnostic is interval coverage rather than a complete rank histogram implementation.

13.4 Simulation-based calibration and rank statistics

In full simulation-based calibration (SBC), one repeats

$$\boldsymbol{\vartheta}_r^* \sim p(\boldsymbol{\vartheta}), \quad \mathcal{D}_r \sim p(\mathcal{D} \mid \boldsymbol{\vartheta}_r^*), \quad \boldsymbol{\vartheta}_{r,s} \sim q(\boldsymbol{\vartheta} \mid \mathcal{D}_r). \quad (13.3)$$

For scalar coordinate k , the rank

$$R_{r,k} = \sum_{s=1}^S \mathbf{1}\{\vartheta_{r,s,k} < \vartheta_{r,k}^*\} \quad (13.4)$$

should be discrete-uniform under an exact posterior and the joint generative model. U-shaped rank histograms suggest underdispersion; central humps suggest overdispersion; slopes suggest bias. Joint or projection-based SBC is needed to probe dependencies that coordinate ranks can miss [12].

Currently implemented: equal-tailed coordinate coverage at three levels, not stored SBC rank histograms. The coverage test catches important over/underconfidence but is less diagnostic about the shape of a failure. Complete rank SBC is a **planned validation extension**; it must not be claimed from the existing coverage image.

13.5 Calibration, sharpness, and simulation budget

Calibration asks whether nominal probabilities have correct repeated frequency. Sharpness asks how concentrated the posterior is, subject to calibration. A prior-like posterior can be calibrated yet scientifically uninformative; an extremely narrow posterior can look impressive while being badly undercalibrated. The desired estimator is as sharp as the data justify, not as sharp as the network can produce.

Simulation-budget sensitivity should repeat training at increasing native group counts while keeping physics and evaluation roles fixed. Stabilizing validation NLL is insufficient if coverage, correlations, tail intervals, or posterior predictive bands still move. The quick and validation profiles provide two compute scales but are not an automated convergence study.

Posterior collapse or false certainty is probed here by coverage, the neural-to-conventional width floor, seed variation, and exact predictive behavior. Failure can mean too little simulation density, an encoder that discarded weak information, flow/optimization error, leakage, or a reference posterior problem. Physical interpretation should wait until these causes are separated.

13.6 Posterior predictive check

Posterior physics samples are exactly reevaluated through PARTONS, their sampled nuisances are applied, and fresh covariance noise is added. This produces replicated observations

$$\mathbf{y}^{\text{rep},s} \sim p(\mathbf{y} | \boldsymbol{\vartheta}^{(s)}), \quad \boldsymbol{\vartheta}^{(s)} \sim q_{\text{ens}}(\boldsymbol{\vartheta} | \mathcal{D}). \quad (13.5)$$

The implementation records how many observed points fall inside pointwise central 90% predictive intervals. This test combines posterior uncertainty, nuisance uncertainty, native predictions, and measurement noise. It should not be confused with simultaneous 90% coverage of an entire 36-vector.

13.7 Exact-native reevaluation

The evaluation profile selects a configured number of posterior samples, sends their 80 physics coordinates through the native bridge, and rejects a campaign if the allowed invalid fraction is exceeded (zero by default). It stores exact CFFs and separately evaluates GPDs over the diagnostic x grid. This is how parameter uncertainty becomes physics-function uncertainty.

The neural flow is therefore not treated as a physics emulator. If a sampled parameter vector causes the native simulator to fail, the failure is recorded as a posterior validity problem rather than replaced by a network prediction.

13.8 What each diagnostic can and cannot establish

Diagnostic	A failure suggests	A pass does not prove
Prior predictive/invalid map	Prior support or simulator validity is poorly matched to the intended experiment	Posterior calibration.
Grouped validation NLL	Overfit, optimization failure, or poor conditional density generalization	Correct credible-set frequency.
Outer-test NLL	Selected estimator does not generalize to locked synthetic groups	Robustness outside the simulator family.
Coverage	Bias or over/underdispersion under the declared joint model	Correct real-data model or every joint dependence.
Posterior predictive	Conditioned model fails to reproduce observed synthetic features	Identifiability of GPD functions.
Conventional comparison	Neural approximation differs from a resolved exact bank posterior	Validity when importance ESS is low.

Diagnostic	A failure suggests	A pass does not prove
Seed ensemble	Optimization instability	Model-family uncertainty.
Named-model holdout	Poor predictive robustness outside the DD training family	Recovery of named-model parameters in DD coordinates.

Chapter 14

Identifiability, shadow directions, and model dependence

14.1 Null and weak directions

Let α parameterize a deformation $F_\alpha = F_0 + \alpha \delta F$. For a finite experimental design with forward map $\mathcal{A}$, an exact observable-null direction satisfies

$$\mathcal{A}[F_\alpha] = \mathcal{A}[F_0] \quad \text{for all admitted } \alpha. \quad (14.1)$$

A weak direction changes predictions by less than the covariance scale. In the exact-null factorized case, the likelihood is independent of α and Bayes' theorem gives

$$p(\alpha \mid \mathcal{D}) = p(\alpha). \quad (14.2)$$

More generally, correlations with other coordinates can preserve a broad ridge rather than a factorized prior marginal. Either behavior is correct: the posterior should retain uncertainty where observations supply no information. Apparent contraction in an exact-null direction indicates prior/model restrictions, leakage, finite neural error, or miscalibration, not successful extraction.

Shadow GPDs exemplify the deconvolution problem: nonzero GPD deformations can have vanishing or extremely small contributions to a specified CFF/order and therefore to chosen observables [17]. Their existence explains why perfect observable agreement need not identify the underlying function. New kinematics, complementary processes/observables, evolution leverage, or external theory constraints may break a weak degeneracy; increasing network capacity alone cannot.

14.2 What shadow tests are actually implemented

The repository contains a separate native Stage 09 identifiability campaign. It composes a GK16 baseline coefficient in $(0.8, 1.2)$ with an installed GPDBMMS21 H up-sea shadow coefficient in $(-1, 1)$, verifies a nonzero GPD deformation, and tests CFF/UU-observable invariance against frozen numerical tolerances on its declared NLO, fixed-scale route. Its grid posterior requires the shadow width to remain essentially prior-wide. This is **currently implemented as a separate physics/identifiability diagnostic**; it is not the 82-dimensional production NPE.

In the production DD simulator, fixed type-level shadow/stress coefficients are injected as part of the truth/simulator family. Only the H/u direction uses the installed BDMMS21 module. The other type/channel directions are project-defined zero-forward-moment DD stress directions and are not proven CFF-null. All are fixed, not among the 82 inferred coordinates. Therefore the production posterior cannot report a learned shadow coefficient, and its historically named `shadow_width_diagnostic` actually describes widths of the 80 inferred DD controls.

A future high-dimensional shadow stress study should promote a verified deformation coefficient to the prior, generate joint simulations across it, and require Eq. (14.2)-like behavior when appropriate, while checking that identifiable baseline directions still contract. Such a study is not completed merely by plotting a fixed injected deformation.

14.3 D-term boundary

The production physics configuration explicitly omits the D-term and the $\tilde{E}$ pion pole. Zero is therefore a model assumption, not a posterior conclusion. The document makes no claim that present data prove a zero D-term, nor that every shadow argument transfers unchanged to D-term physics. Introducing it would add a different functional contribution and requires a new parameterization, prior, corpus identity, and validation campaign.

14.4 Six layers of model dependence

It is useful to separate what a result combines:

1. *Experimental information*: kinematics, observed values, and the uncertainty/nuisance model.
2. *Prior information*: scalar supports and any assumed correlations.
3. *GPD representation*: DD profile, endpoint form, omitted terms, flavor basis, and fixed stress settings.
4. *Perturbative physics*: LO coefficient/evolution order, $\overline{\text{MS}}$ scheme, scales, and fixed three-flavor treatment.
5. *Finite simulation*: which part of the prior-predictive joint was represented by the corpus.
6. *Neural approximation*: encoder compression, MAF family, optimization, and ensemble selection.

The posterior is conditional on all six layers; only the first is directly experimental.

Switching from DDs to conformal moments or another family can change skewness, analytic structure, constraints, and which directions are easy to express. Even with identical data, priors that look “similar” in their native coordinates need not induce similar priors over GPD functions. Comparing model families is scientifically valuable, but should use prior-predictive matching where possible and report between-family spread separately from within-family credible bands.

Currently implemented: one production DD NPE; native conformal/DD probes; post-training GK/VGG holdouts. **Not implemented:** Bayesian model averaging over DD and conformal families, perturbative-order marginalization, or a parameterization-independent posterior.

Chapter 15

The conventional posterior comparison

15.1 Prior-bank importance sampling

The noise-free corpus is already a bank of physics draws from the prior. For each bank vector, the comparison draws nuisance values from their priors, applies Eq. (4.5), and evaluates the Gaussian residual using the exact native prediction. Since the proposal is the joint prior, the unnormalized importance weight is simply

$$\tilde{w}_j = p(\mathbf{y}_{\text{obs}} | \boldsymbol{\vartheta}_j) \propto \exp\left(-\frac{1}{2} \mathbf{r}_j^T C^{-1} \mathbf{r}_j\right). \quad (15.1)$$

After stable subtraction of the maximum log likelihood, $w_j = \tilde{w}_j / \sum_l \tilde{w}_l$. The effective sample size is

$$\text{ESS} = \frac{1}{\sum_j w_j^2}. \quad (15.2)$$

Weighted proposals are resampled to form the conventional reference sample.

15.2 Why ESS is decisive

In high dimension, a small number of prior draws may carry almost all likelihood weight. Then re-sampling duplicates a few points, tail estimates are poor, and distances to NPE are not trustworthy. The configured minimum ESS is 80. Failure means the conventional reference is under-resolved; it does not validate or invalidate NPE by itself.

15.3 Comparison metrics

The two posterior samples are standardized by conventional marginal means and standard deviations. The framework computes:

- one-dimensional Wasserstein distances for every parameter, divided by the conventional marginal standard deviation;
- a sliced Wasserstein distance averaged over 128 random projections in the standardized 82-dimensional space; and

- a false-certainty diagnostic requiring every neural central-90% physics interval to be at least 0.85 times the corresponding conventional interval width.

These metrics compare two finite approximations to the same declared posterior. Agreement is a closure check, not evidence that the DD family or synthetic covariance describes nature.

Chapter 16

Fresh model holdouts and real-data boundary

16.1 Named native-model holdouts

Only after synthetic closure gates pass may a frozen NPE be tested on fresh PARTONS truth from GK11, GK16, GK19, and VGG99. These named-model outputs are forbidden from corpus generation, optimization, training, or ensemble selection. Hashes protect earlier artifacts during the holdout operation.

The holdout probes predictive robustness outside the training DD family. It does not measure recovery of an 80-component DD truth, because the named models do not possess the same coordinate vector. A mismatch is a scientific finding about robustness; tuning repeatedly against it destroys its blind status.

16.2 Why real data are quarantined

The installed database and measurement catalog can help inspect kinematic coverage, but real values and uncertainties do not enter the likelihood or network. A credible real-data extraction would additionally require audited observable definitions and signs, units, radiative and acceptance conventions, complete covariance or a defensible missing-correlation policy, dataset-level nuisances, and publication/data-use review. The present software refuses to label a diagnostic overlay as a fit.

Part V

Using and auditing the framework

Chapter 17

Operational workflow

17.1 A clean campaign

From the repository root, a typical explicit workflow is:

```
./dvcs init my-study
./dvcs corpus-create my-study my-corpus --profile validation --shard-size 256
./dvcs corpus-plan my-study my-corpus
./dvcs corpus-generate my-study my-corpus
./dvcs corpus-verify my-corpus --deep
./dvcs selection-create my-study my-corpus baseline --profile validation
./dvcs train my-study --profile validation \
  --corpus my-corpus --selection baseline
./dvcs evaluate my-study --profile validation
./dvcs compare my-study --profile validation
./dvcs plot my-study --profile validation
./dvcs holdout my-study --profile validation
```

Project creation syntax and optional data-source controls are documented by `./dvcs help` and the CLI reference. The important methodological order is corpus verification, frozen selection, training, synthetic evaluation, conventional comparison, plotting, and only then fresh holdout.

17.2 What each expensive step computes

Action	Methodological effect
corpus-generate	Draws prior physics groups and runs exact native PARTONS simulations; no neural training and no measurement noise.
selection-create	Freezes group ownership for training, internal validation, and outer test; no simulation.
train	Materializes deterministic nuisance/noise replicas, builds contexts, and fits candidate conditional flows.

Action	Methodological effect
optimize	Repeats validation-only one-seed training across selected neural controls; never selects with outer test or real data.
evaluate	Runs outer-test NLL, repeated coverage, posterior sampling, exact native posterior reevaluation, and posterior predictive checks.
compare	Builds the exact-bank importance posterior and compares it with the neural ensemble; no retraining.
holdout	Tests frozen checkpoints against fresh named native models after closure; protected earlier outputs cannot change.
compare-real	Produces a read-only diagnostic overlay; not a likelihood, reweighting, training step, or posterior update.

17.3 CPU and GPU roles

Corpus generation and exact reevaluation are CPU-native and parallelize with isolated PARTONS worker processes because native thread safety is not assumed. NPE training and sampling are PyTorch work and benefit from a GPU. On multiple GPUs, independent ensemble members or Optuna trials are sharded; the implementation is single-node and does not distribute one model across nodes. A larger allocation does not compensate for insufficient simulation coverage or an invalid statistical design.

Chapter 18

Reading results without overclaiming

18.1 A hierarchy of questions

Interpret a campaign in this order:

1. Did every corpus and selection integrity check pass?
2. Were native invalid draws and exact posterior invalid draws acceptable?
3. Did training converge without a validation/test generalization gap?
4. Is repeated coverage compatible with nominal coverage at its finite precision?
5. Does the posterior predictive distribution cover the synthetic data?
6. Is the conventional ESS adequate, and only then do posterior distances pass?
7. Do exact-native CFF and GPD bands have physically interpretable shapes?
8. After freezing all choices, what happens on fresh named-model holdouts?

Skipping earlier questions makes later plots difficult to interpret.

18.2 Broad posterior directions

Broad marginals are expected when observables do not identify a direction. They are not automatically a training failure. Compare posterior width with the prior and conventional reference, inspect correlations, and look at predictive quantities. Conversely, a very narrow neural interval can be a failure if coverage is low or it is substantially narrower than a reliable conventional posterior.

18.3 Correlations and marginal plots

A one-dimensional histogram integrates over 81 other coordinates. Two different joint posteriors can have similar marginals, and strong curved degeneracies can make a marginal median physically unrepresentative. Use joint projections, CFF/GPD reevaluations, sliced distances, and predictive checks. Avoid interpreting 82 separate 90% intervals as a simultaneous 90% region.

18.4 Injected truth

Injected truth is known only because the study is synthetic. In one noise realization it need not be the posterior mode or median. A method that always pulls the answer visually toward truth could be underestimating uncertainty. Across repeated joint draws, calibrated intervals should cover at their nominal rates within sampling uncertainty.

18.5 Failure modes and their likely meanings

Symptom	Interpretation to investigate
Training NLL falls while validation NLL rises	Capacity, optimization, or finite-corpus overfitting; use the frozen validation rule, not outer-test tuning.
Outer-test NLL is much worse	Group-level generalization failure or insufficient prior-predictive coverage.
Undercoverage	Posterior is too narrow, flow/training approximation is poor, or the context fails to encode relevant information.
Overcoverage	Posterior may be too diffuse; finite trial count can also make the deviation noisy.
Poor predictive coverage	Posterior, nuisance/noise model, or native prediction chain does not reproduce the conditioned synthetic dataset.
Low conventional ESS	Prior importance bank is too sparse for this dataset; do not treat its distances as authoritative.
Invalid exact posterior samples	Learned mass has entered a native-invalid region despite nominal bounded support.
Named-model holdout mismatch	Lack of robustness beyond the training DD family; report it without adapting to the same holdout.
Apparently precise parameters but broad physics curves	Marginal summaries or correlations are being misread; inspect exact joint reevaluations.

Chapter 19

Limitations and principled extensions

19.1 Assumption register

Assumption	Current choice	Possible consequence
GPD representation	Independent five-control DD block for each of four GPDs and four channels	Smoothness/skewness/model constraints can exclude other functions and understate between-family uncertainty.
Input scale	$Q_0^2 = 1 \text{ GeV}^2$	Parameters and priors are defined at this scale; changing it changes their interpretation.
Evolution/order	LO, $\overline{\text{MS}}$, fixed $n_f = 3$	Missing-order, scale, threshold, and heavy-flavor uncertainty is absent.
Omitted terms	No D-term, $\tilde{E}$ pion pole, or heavy-quark input	A narrow posterior cannot be evidence that these contributions vanish.
Kinematic domain	Validated fixed-target envelope and $0 < y < 1$; upward Q^2 evolution only	Calibration is not established outside the envelope.
Prior	Independent bounded uniforms for 80 controls and normal nuisances	Independence/support choices can dominate weak directions.
Experimental model	Fixed truth-based Gaussian covariance and two linear normalization nuisances	Synthetic closure does not establish real systematic or non-Gaussian noise fidelity.
Simulator fidelity	Installed native module chain and numerical settings	Unmodeled native/systematic theory error is not posterior uncertainty.

Assumption	Current choice	Possible consequence
Simulation budget	2,048/4,096 native groups in standard profiles	Sparse coverage in 82 dimensions can bias the learned density; names do not certify adequacy.
Amortization	Single-round prior-predictive NPE	Valid only over the trained joint distribution; no automatic out-of-distribution correction.
Posterior family	Six-transform conditional affine MAF ensemble	Finite flow capacity/optimization can miss modes or tails.
Dataset compression	Fixed-count DeepSets with mean pooling to 96 features	Pooling may discard pairwise or weak-direction information.
Validation	One frozen grouped split and finite coverage trials	Monte Carlo resolution and split dependence remain; K -fold validation is absent.

Every row is part of the meaning of a result. A proposed extension should state which row it changes and repeat the affected validation; it should not be described as a mere neural-network upgrade.

19.2 Current limitations

The central statistical limitations are high posterior dimension relative to the default measurement set; finite prior-predictive simulation density; one frozen split rather than K -fold studies; materialization rather than lazy streaming; finite coverage trials; an importance reference that can collapse; and calibration only inside the declared simulator envelope. Multi-GPU execution is single-node. Real-data inference is deliberately unavailable.

The physics family is also restricted: full-independent DD controls at one input scale, fixed-three-flavor LO evolution, a fixed native module chain, fixed shadow/stress settings, and no general selector for arbitrary twist, order, scheme, threshold, or process modules. The prior itself is part of this family.

19.3 How to extend without invalidating inference

An extension should first define a versioned generative contract, then update the native simulator and corpus identity, token semantics, priors/transforms, network dimensions, artifact schemas, and validation campaign together. It must receive new coverage and predictive validation. Merely widening a tensor or reusing an old checkpoint is not valid.

Examples include adding a new observable after native convention audits, introducing dataset-specific nuisances, learning a different GPD representation, adding heavy-flavor evolution, or moving to real data with a complete experimental likelihood. Each changes the probability model and therefore the meaning of the posterior.

Appendix A

Exact posterior parameter contract

The first 80 names are generated in the nested order

$$F \in \{H, E, H\text{tilde}, E\text{tilde}\}, \quad (\text{A.1})$$

$$p \in \{u, d, s, \text{gluon}\}, \quad (\text{A.2})$$

$$s \in \{\text{normalization}, a, c, \text{profile_b}, t_slope_GeV_minus2\}. \quad (\text{A.3})$$

The exact string is F_p_s . Thus the vector starts

```
H_u_normalization, H_u_a, H_u_c, H_u_profile_b,
H_u_t_slope_GeV_minus2, H_d_normalization, ...
```

continues through every H , E , $H\text{tilde}$, and $E\text{tilde}$ channel, and ends with

```
Etilde_gluon_normalization, Etilde_gluon_a, Etilde_gluon_c,
Etilde_gluon_profile_b, Etilde_gluon_t_slope_GeV_minus2,
eta_global_normalization, eta_lu_normalization
```

Field	Physical support	Latent map
normalization	$(-4, 4)$	probit of scaled uniform
a	$(0.1, 1)$	probit of scaled uniform
c	$(2, 6)$	probit of scaled uniform
profile b	$(1, 4)$	probit of scaled uniform
t slope B	$(0, 2) \text{ GeV}^{-2}$	probit of scaled uniform
η_g, η_{LU}	$\mathbb{R}$	identity

Appendix B

Exact context contract

Index	Feature name	Meaning
1	x_b_over_0p3	$x_B/0.3$
2	minus_t_GeV2_over_0p2	$-t/(0.2 \text{ GeV}^2)$
3	q2_over_q0_squared	$Q^2/(1 \text{ GeV}^2)$
4	beam_energy_GeV_over_12	$E/(12 \text{ GeV})$
5	sin_phi	$\sin \phi$
6	cos_phi	$\cos \phi$
7	observable_is_uu	UU cross-section one-hot bit
8	observable_is_lu_difference	LU-difference one-hot bit
9	observable_is_beam_charge_asymmetry	beam-charge one-hot bit
10	observable_is_beam_spin_asymmetry	beam-spin one-hot bit
11	observable_is_target_spin_asymmetry	target-spin one-hot bit
12	observable_is_double_spin_asymmetry	double-spin one-hot bit
13	observed_normalized	normalized observed value y_i
14	marginal_sigma_normalized	$\sqrt{C_{ii}}$
15	symmetric_whitened_observation_over_10	$(C^{-1/2}\mathbf{y})_i/10$
16	global_normalization_response	$r_g = 0.02$
17	lu_normalization_response	$r_{LU,i} \in \{0, 0.03\}$
18	point_mask	one for every present token

The nine global features, in order, assert: full-independent four-GPD flavor DD representation; $Q_0^2/(1 \text{ GeV}^2)$; LO coefficient function; configured evolution; native datum- Q^2 evolution; full covariance; all four twist-2 GPD types; native six-observable route; and model mask. With 36 tokens, the flattened context width is $36 \times 18 + 9 = 657$ before DeepSets reshaping.

Appendix C

Default neural and campaign controls

Control	Default	Effective meaning
Estimator	<code>sbi.NPE</code>	single-round neural posterior estimation
Flow	<code>zuko_maf</code>	conditional masked autoregressive affine flow
Point hidden/layers	128 / 3	shared SiLU token MLP
Dataset hidden/layers	128 / 2	post-pooling SiLU MLP
Embedding features	96	flow-conditioning dimension
Point LayerNorm	false	optional first-layer normalization disabled
Flow hidden features	96	six ReLU hidden layers of this width per current masked conditioner
Flow transforms	6	alternating-order autoregressive transforms
Bins	8	ignored for Zuko MAF by <code>sbi 0.26.1</code>
Parameter standardization	independent	estimator-side latent standardization
Context standardization	none	explicit token scales retained
Batch size	256	training rows per mini-batch
Learning rate	5×10^{-4}	<code>sbi</code> optimizer control
Internal-validation fraction	0.1 of groups	about 8% overall
Quick corpus/replicas	2,048 / 2	4,096 realized rows
Validation corpus/replicas	4,096 / 4	16,384 realized rows

Control	Default	Effective meaning
Quick epochs/patience	100 / 20	maximum / no-improvement stopping
Validation epochs/patience	200 / 25	maximum / no-improvement stopping
Quick candidate/active ensemble	2 / 2	seeds 51001–51002
Validation candidate/active ensemble	5 / 3	select three by validation NLL
Outer-test fraction	0.2	group-disjoint and locked

Appendix D

Frozen validation gates

The shipped configuration declares the following campaign thresholds. They are acceptance criteria chosen for this workflow, not universal constants:

Quantity	Threshold
Minimum conventional ESS	80
Maximum ensemble sliced Wasserstein	0.45
Maximum marginal Wasserstein / conventional SD	0.60
Maximum outer-test NLL minus validation NLL	0.75
Maximum coverage deviation in binomial SE units	3.5
Minimum pointwise posterior-predictive 90% coverage	0.75
Minimum neural/conventional central-90% width ratio	0.85
Maximum exact-reevaluation invalid fraction	0

Passing means only that the predeclared checks passed for the recorded campaign. Inspect each metric and its Monte Carlo resolution; do not reduce the scientific conclusion to one Boolean.

Appendix E

Implementation map

Path	Responsibility
<code>configs/inference/stage10_pseudodata_workflow.json</code>	Priors by support, kinematics, observables, covariance, profiles, context, network, Optuna, seeds, and gates.
<code>configs/physics/stage10_pseudodata_systematics_v1.json</code>	Native DD, evolution, CFF, process, observable, scale, and domain contract.
<code>src/extract_dvcs_cff/inference/stage10.py</code>	82-coordinate transforms and prior, full-covariance context, grouped NPE data loaders, density-estimator factory, state-only loading.
<code>src/extract_dvcs_cff/inference/deepsets.py</code>	Shared point MLP, mean pooling, global-feature concatenation, dataset MLP.
<code>src/extract_dvcs_cff/workflows/pseudodata.py</code>	Covariance/noise generation, realization, training, ensemble, coverage, exact reevaluation, conventional comparison, holdout, and reports.
<code>src/extract_dvcs_cff/optimization/stage10.py</code>	Persistent validation-only Optuna study.
<code>src/extract_dvcs_cff/corpus.py</code>	Reusable sharded exact-native corpus, selection, integrity, transfer, and materialization contracts.
<code>cpp/src/partons_bridge.cpp</code>	Versioned native request validation and PARTONS service execution.
<code>docs/KNOWN_LIMITATIONS.md</code>	Current verified/unverified claim boundary.
<code>provenance/dependencies.lock.json</code>	Pinned dependency authority.

Appendix F

Glossary

Amortized inference

Up-front training produces a conditional estimator that can be reused for compatible datasets.

Calibration

Agreement between stated posterior probabilities and frequencies over repeated experiments generated by the declared model.

Conditional density estimator

A normalized model for a distribution of parameters whose shape depends on an input dataset.

Context

The flattened token features plus global contract features provided to the DeepSets encoder.

Corpus group

One 80-component physics draw and all of its noise-free native outputs.

DeepSets

Shared token transformation followed by symmetric pooling and a dataset-level transformation.

Exact-native

Recomputed through the C++ bridge and PARTONS, without a Python physics surrogate.

Flow

An invertible transformation that turns a simple base density into a flexible density with a tractable Jacobian.

MAF

Masked autoregressive flow; masks give a triangular dependency and tractable density.

NPE

Neural posterior estimation, trained by conditional log density on joint simulator draws.

Outer test

Grouped synthetic pairs kept out of training, early stopping, Optuna, and ensemble selection.

Posterior predictive

Distribution of replicated observations after drawing parameters from the posterior and observations from the simulator.

Realization

Nuisance/noise replicas and neural tensors deterministically built from a corpus and selection.

SBI Bayesian inference based on simulated parameter/data pairs rather than differentiating or repeatedly evaluating an internal simulator formula.

Selection

Immutable assignment of corpus group indices to train, internal validation, and outer test.

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Document maintenance rule

When the posterior dimension, token contract, network builder, **sbi**/Zuko version, split policy, generative model, or validation gates change, this document must be reviewed in the same commit. A methods book that describes a previous estimator is worse than a short document because it creates false confidence.